

Predict Students Dropout



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Overview

Why It Matters (Portugal)

According to DGEEC (Feb. 2024):

- 11% of 1st-year undergraduates in 2021/2022 dropped out — they were no longer enrolled nor had graduated one year later.

Highlighting the real and urgent need for effective intervention strategies in Portuguese higher education.

State of the Art

- Deb et al. (2024): RF model on Bangladeshi data with a 78% F1-score.
- Karimi-Haghighi et al. (2022): incorporated fairness in dropout prediction.

Most studies using this dataset were conducted outside Portugal (e.g., Kim et al. in the U.S.).

This study brings a local perspective, using real national data and exploring actionable strategies for early identification.

Data Set

Consists of 4,424 student records.

35 attributes that cover a wide covering a wide range of academic, demographic, and socioeconomic factors

The target variable is categorical and originally includes three classes: Graduate, Enrolled and Dropout.

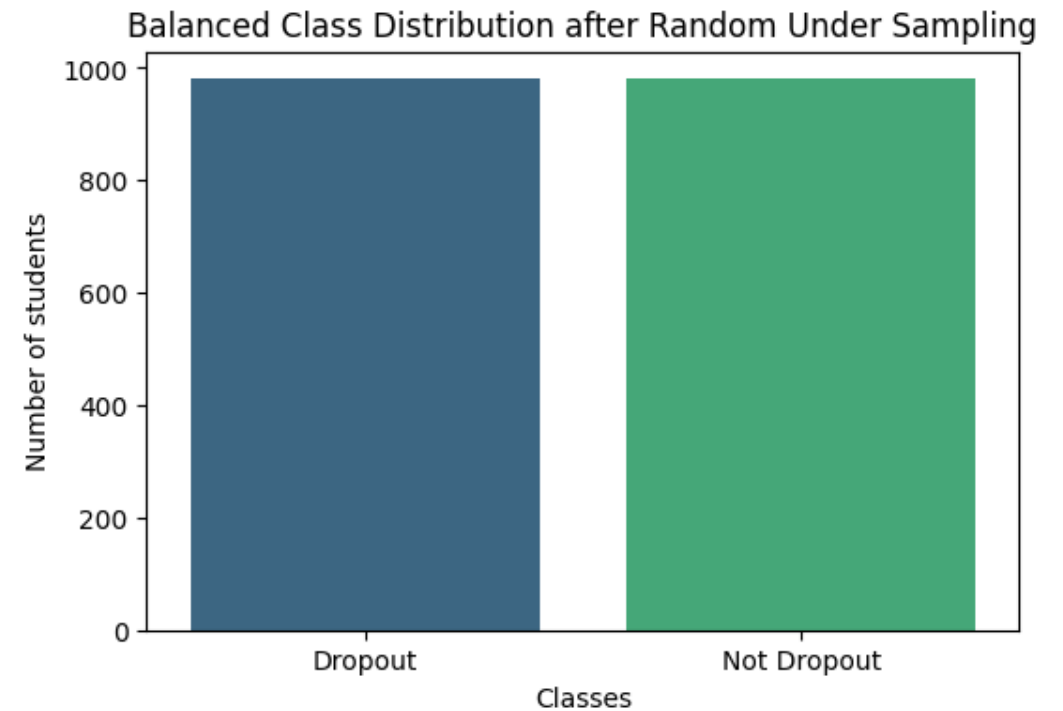
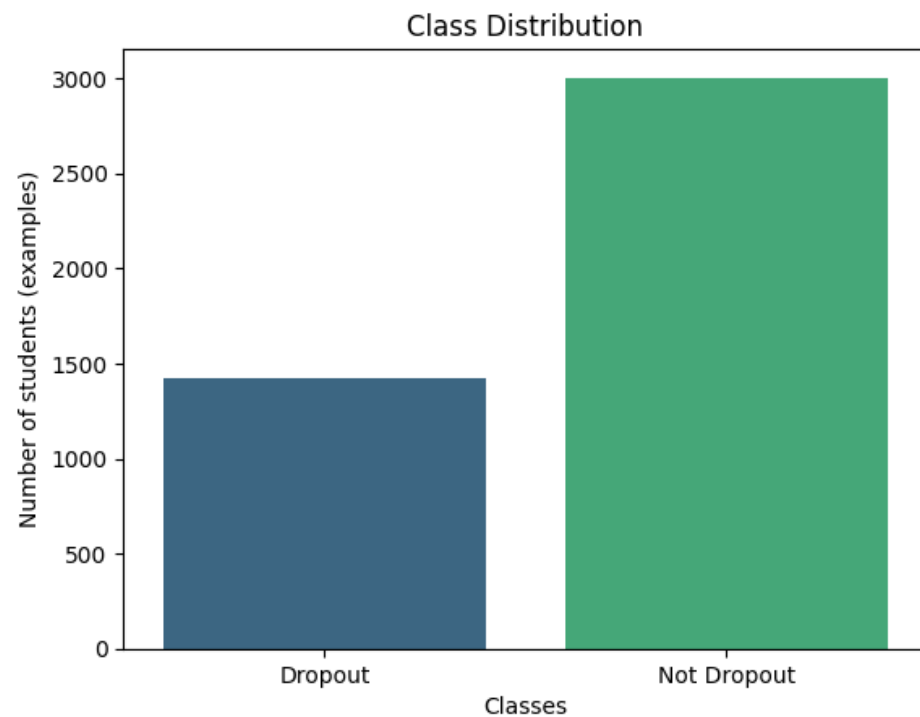
Developed by four professors of Instituto politécnico de Portalegre: Valentim Realinho, Jorge Machado, Luis Batista, and Mónica V. Martins.

Pre-Processing

- Converting the labels to fit the desired problem
- Splitting the data (70-15-15)
- Data balancing
- Feature scaling and Standardization

Data balancing

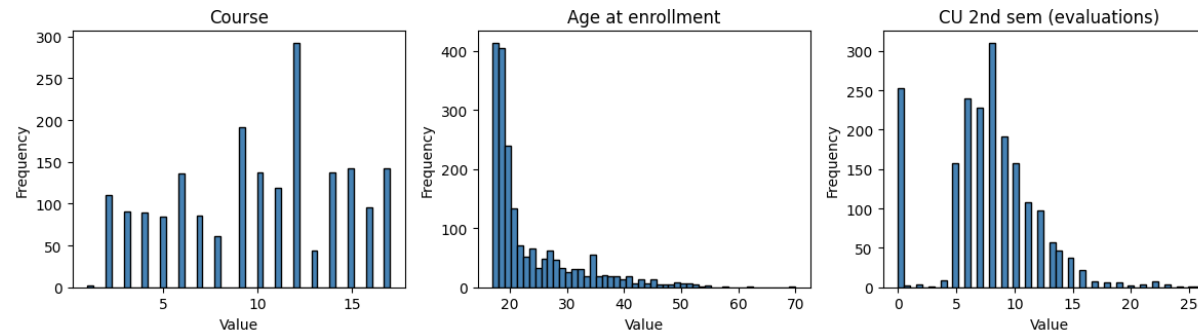
| Pre-Processing



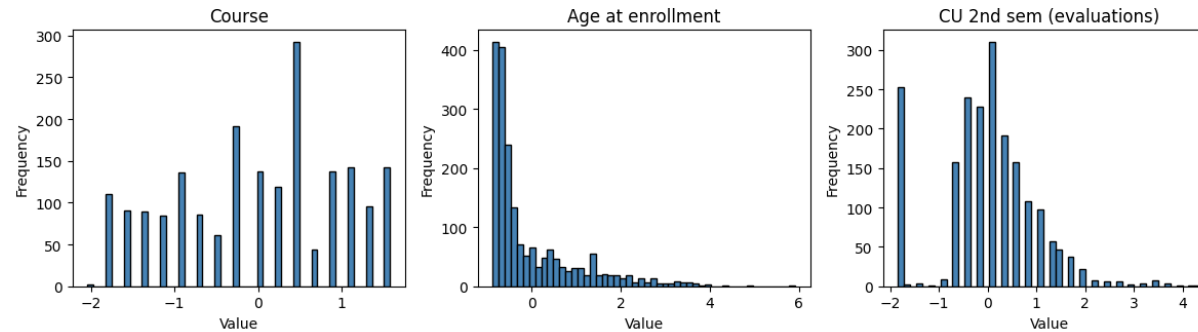
Feature Scaling and Standardization

Pre-Processing

Selected Features Distributions before Standardization



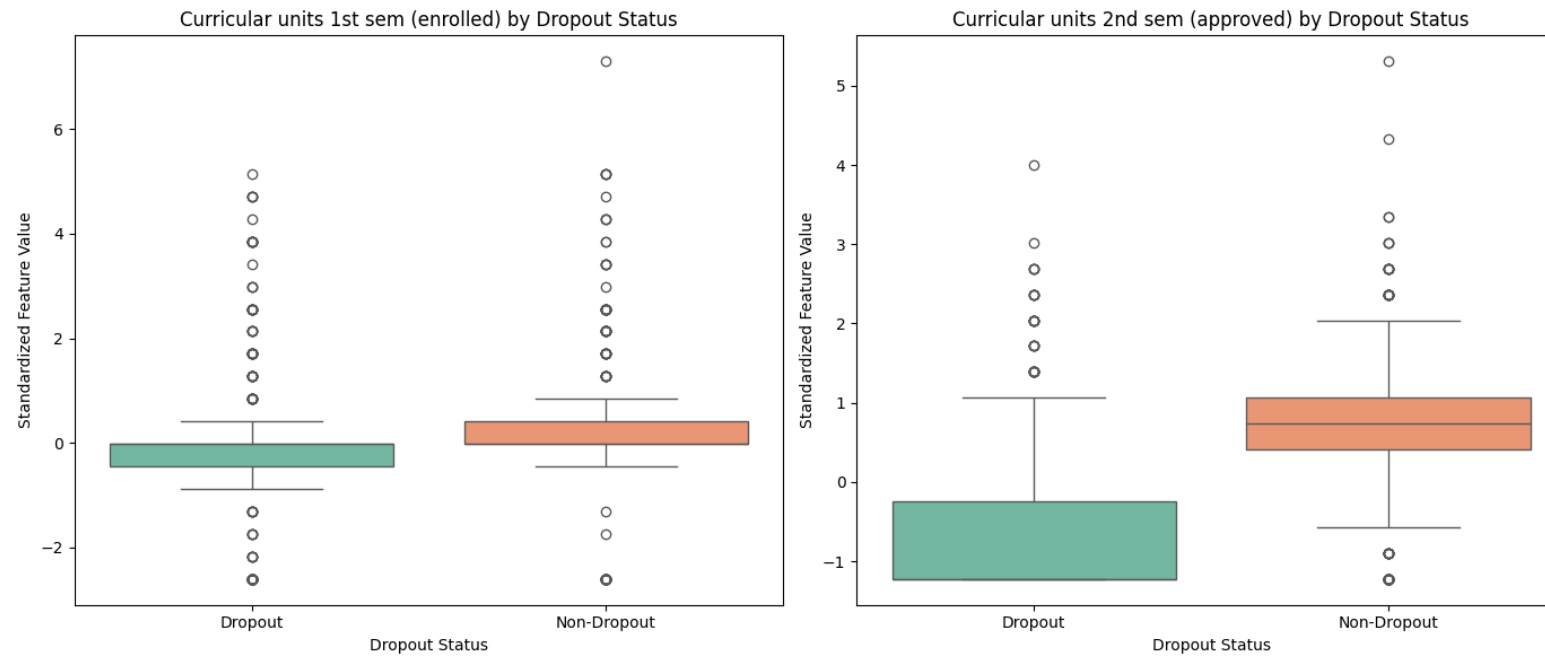
Selected Features Distributions after Standardization



Standardization does not bound values to a specific range, which in some cases may be a problem but, on the other hand, is much less affected by outliers

Data visualization

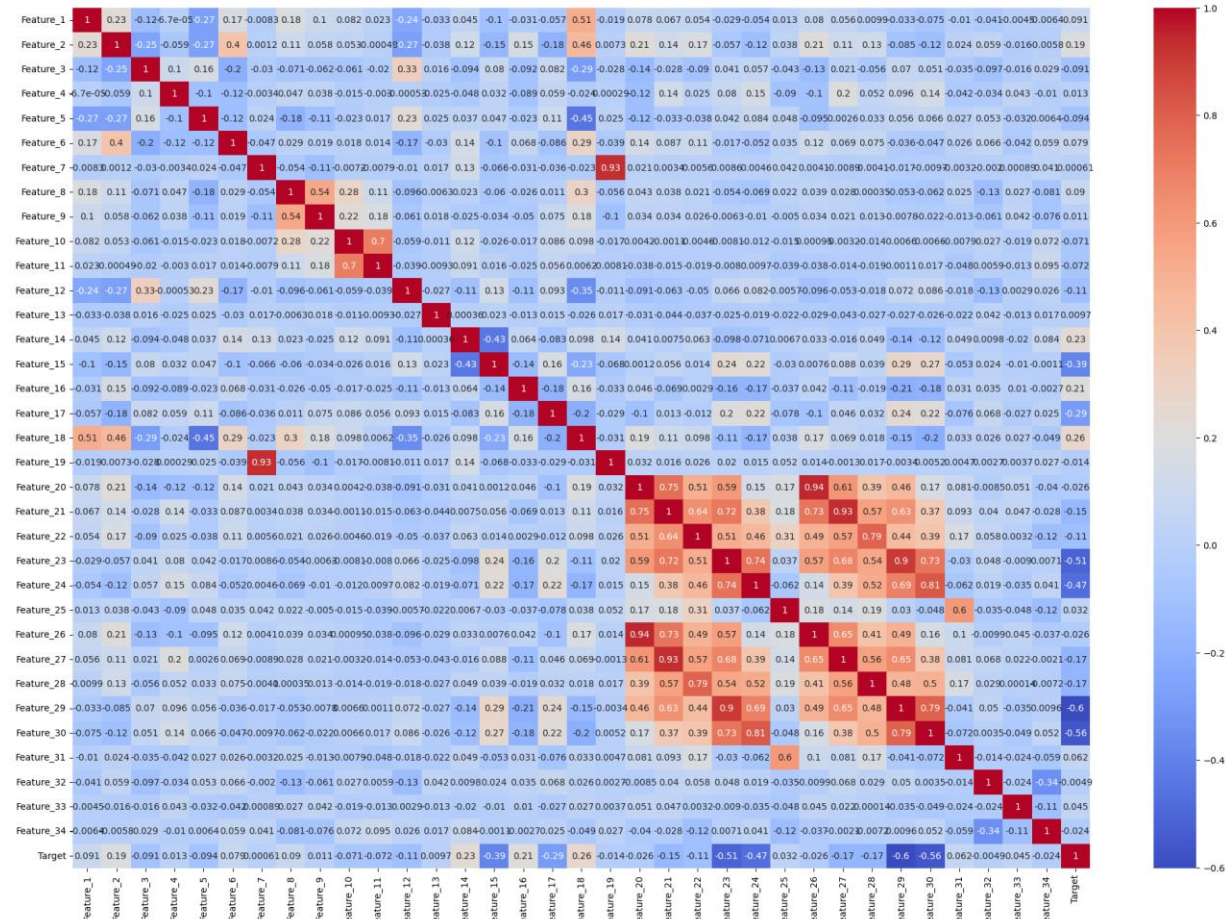
| Box Plots



Used to visualize the influence of a feature in the classes, if the separation is minimum then so is its influence.

Data visualization

| Correlation Matrix



Models Used

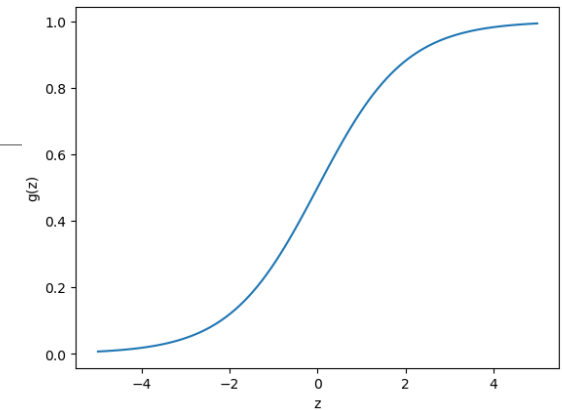
- Logistic Regression
- Support Vector Machine

All models used were implemented using the scikit learning library.

Logistic Regression

Logistic function (sigmoid)

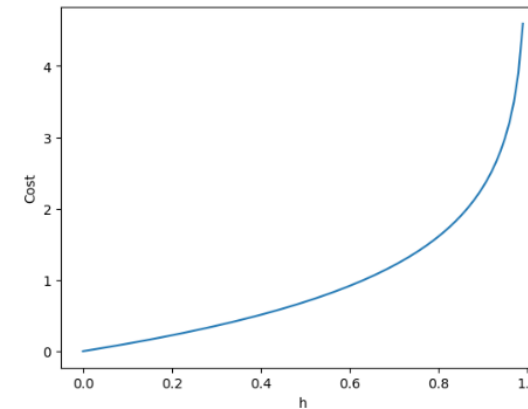
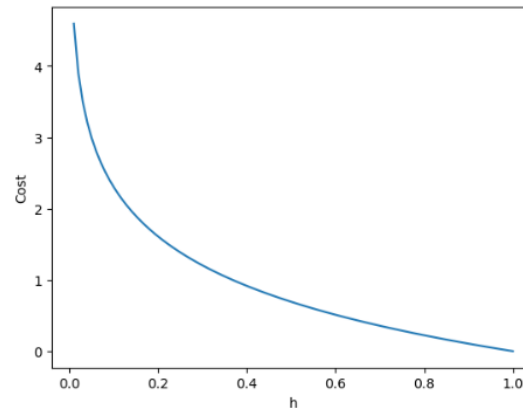
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$



Cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

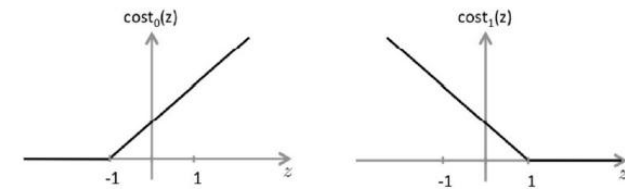


Models the probability that a given input belongs to a given class and tries to find the best decision boundary that maximizes the chance of correctly classifying the data.

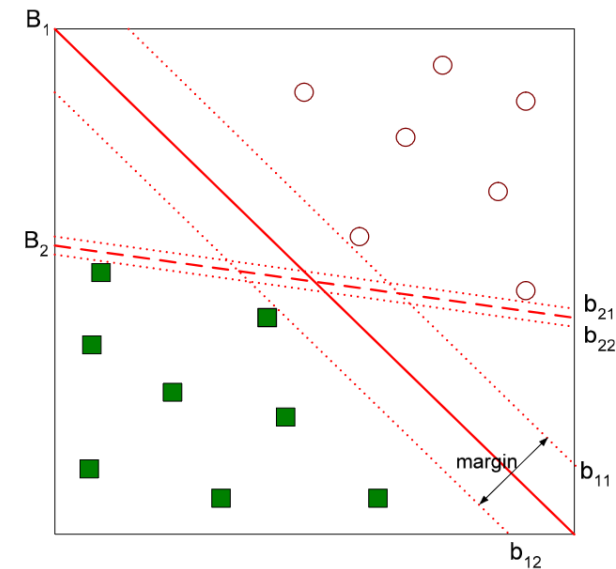
Support Vector Machine

Regularized Cost Function

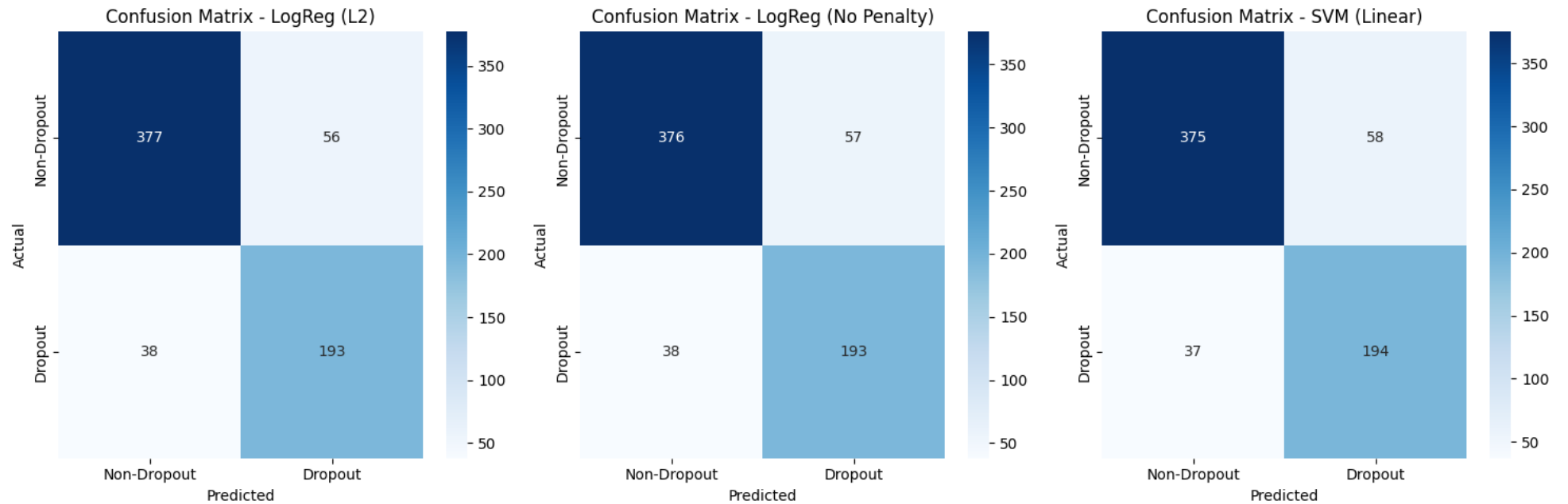
$$\min_{\theta} C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$



SVM tries to find the hyperplane that separates the two classes with the largest possible margin between classes.



Baseline Evaluation



LogReg (L2) – default hyperparameter in scikit learn ($C = 1$) where $C = 1/\text{lambda}$

Baseline Evaluation

Table 1. Performance on the training set for the baseline models.

Model	Accuracy	Precision	Recall	F1 Score	AUC
LogReg (L2)	0.8582	0.8799	0.8296	0.8540	0.9288
LogReg (No Penalty)	0.8561	0.8785	0.8265	0.8517	0.9288
SVM (Linear)	0.8602	0.8854	0.8276	0.8555	0.9271

Table 2. Performance on the validation set for the baseline models.

Model	Accuracy	Precision	Recall	F1 Score	AUC
LogReg (L2)	0.8584	0.7751	0.8355	0.8042	0.9205
LogReg (No Penalty)	0.8569	0.7720	0.8355	0.8025	0.9207
SVM (Linear)	0.8569	0.7698	0.8398	0.8033	0.9173

Feature Selection

Lasso regression

- 21 features selected with $C=0.1$

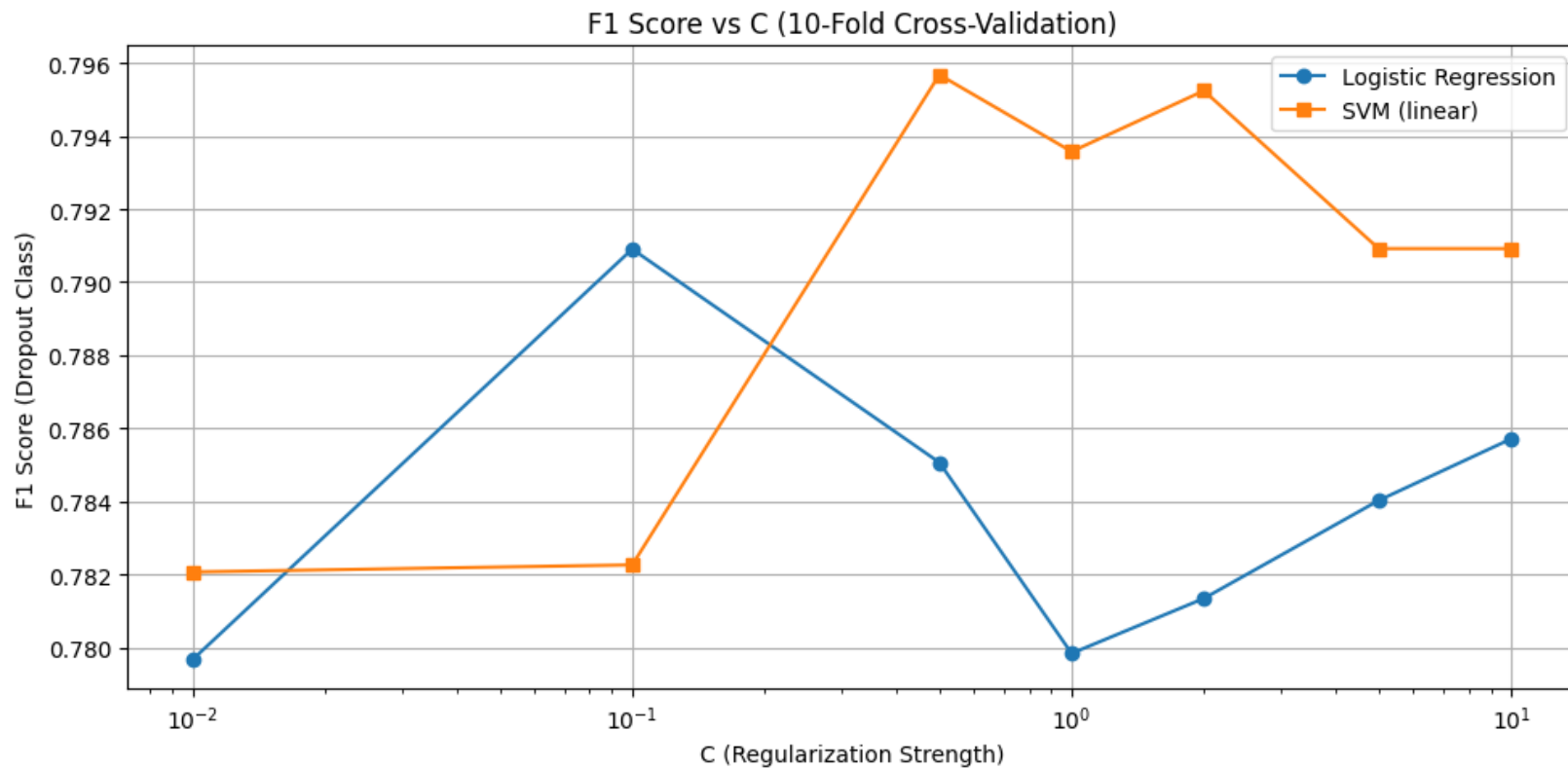
Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8584	0.7751	0.8355	0.8042
SVM (linear)	0.8569	0.7698	0.8398	0.8033

Tabela 1: Performance on Full Set of Features

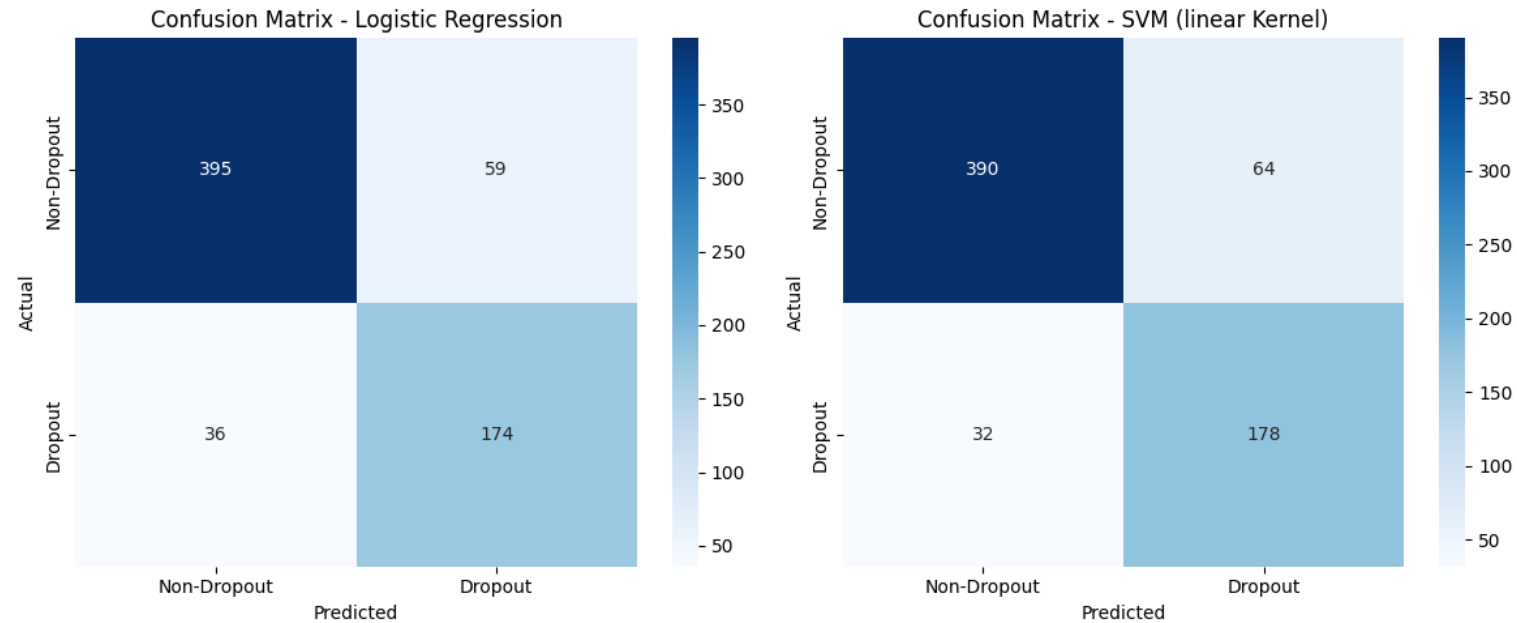
Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8554	0.7711	0.8312	0.8000
SVM (linear)	0.8539	0.7702	0.8268	0.7975

Tabela 2: Performance on Lasso Set of Features

Hyperparameter Tuning



Results



Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.8569	0.7468	0.8286	0.7856	0.9192
SVM (linear Kernel)	0.8554	0.7355	0.8476	0.7876	0.9169

Conclusion

- ✓ Machine learning models can effectively predict student dropout risk.
- ✓ Both Logistic Regression and SVM performed well — SVM had higher recall.
- ✓ High recall is crucial for early intervention and student support.
- ✓ Feature selection with Lasso simplified the model but offered no significant gain.
- ✓ Data-driven approaches can help institutions reduce dropout rates.

Thank you for your
attention! 